

Total No. of Questions :10]

SEAT No. :

P2616

[Total No. of Pages :3

[5153] - 592

T.E. (I.T.)

THEORY OF COMPUTATION

(2012 Pattern) (Semester - I) (End-Semester) (314442)

Time : 2½Hours]

[Max. Marks :70

Instructions to the candidates:

- 1) Neat diagrams must be drawn wherever necessary.
- 2) Figures to the right side indicate full marks.
- 3) Assume suitable data if necessary.

- Q1)** a) Construct FA that accepts even number of zeros & odd number of ones. [6]
- b) Write formal definition of regular expression with suitable example. State Arden's theorem and its use. [4]

OR

- Q2)** a) Construct a Moore machine to find out the residue-modulo-3 for binary number. [6]
- b) Define regular sets. List out closure properties of regular sets. [4]
- Q3)** a) Define the following and give appropriate examples [4]
- i) Derivation tree
 - ii) Context free grammar
- b) Convert right linear grammar to its equivalent left linear grammar. [6]

$S \rightarrow bB$

$B \rightarrow bC$

$B \rightarrow aB$

$B \rightarrow b$

$C \rightarrow a$

OR

P.T.O.

Q4) a) Construct a DFA equivalent to the following grammar [6]

$S \rightarrow S10|0$

b) Write a short note on the applications of CFG. [4]

Q5) a) Design a PDA that checks wellformedness of parentheses. Simulate PDA for $(() (()))$. [8]

b) Define and compare DPDA and NPDA. Justify that NPDA is more powerful than DPDA. [8]

OR

Q6) a) Construct a PDA for the language generated by the following grammar [8]

$S \rightarrow aB \mid bA$

$A \rightarrow bAA \mid a \mid aS$

$B \rightarrow b \mid bS \mid aBB$

b) Define post machine. Compare FA, PDA and post machines. [8]

Q7) a) Write a short note on: [8]

i) Church Turing Hypothesis

ii) Post correspondence problem

b) Design a Turing Machine to recognize the language $L = \{1^n 2^n 3^n \mid n \geq 1\}$. Simulate TM for "112233". [10]

OR

Q8) a) Design a Turing machine that accepts $L = \{0^n 1^n \mid n \leq 1\}$. Simulate TM for "000111". [10]

b) Explain the following for a TM [8]

i) Power of TM over finite state machine

ii) Universal TM

- Q9) a)** Write a short note on decidable problems concerning [8]
- i) Context free languages
 - ii) Turing machines
- b) What is reducibility? What are undecidable problems? Describe at least four undecidable problems in case of CFGs. [8]

OR

- Q10)a)** Describe post correspondence problem. PCP is an unsolvable problem. Justify. [8]
- b) What are recursive and recursively enumerable languages? What is the relation between them? [8]
